



POLITÉCNICA

Curso 2023-24
Seguridad de la Información
Block III



Objectives

Prepare to the exam number III:

- Unit 6: Digital Signature
- Unit 7: Digital Certificates

Exercises

Exercise 1

The famous Spanish entity “Banco Forrado” (in English: “Loaded bank”) has a website. It communicates with users through a protocol called “MiniSSL”. As a result of this protocol, the bank sends the customer its digital certificate, issued by the authority Don Corleone. The content of the certificate is:

Issued by: Don Corleone
Issued for: Banco Forrado
Valid from: 1/1/2023
Public Key: 7
Mod: 403
Signature Algorithm: RSA over DIM5
Signature: 79

By downloading Don Corleone's certificate, you get the public key of that authority: $e = 5$, $n = 391$.

In addition, the **DIM5** hash function (*which does not exist, it is made-up for this exercise!!!*) creates an 8-bit hash of a text document, as follows:

- The first two bits refer to the number of lines of text mod 4.
- The next two bits refer to the length of the longest line (counting white space) mod 4.
- The next two bits represent the first integer that appears in the document, mod 4 (and if there is none: 00).
- The last two bits count the occurrences of the letter 'a' (lowercase) mod 4.

Questions

1. Calculate the DIM5 hash of the certificate above, knowing that it has 6 lines, the longest line is 31 characters, the first integer is 1, and the number of symbols 'a' is 8.
2. Verify the signature that appears on the certificate
3. Does DIM5 have weak collision resistance? Clearly motivate your answer.

Exercise 2

A University (Universidad a Distancia) is identified online through the "Simplified-SSL" protocol, with a digital certificate issued by the TrustMePlease authority. The content of this certificate is:

Issuer: TrustMePlease
Owner: Universidad a Distancia
Valid from: 1/1/2023
Validez until: 1/1/2024
Public Key: 65537
Modulus: <i>(number of 256 bytes)</i>
Signature Algorithm: RSA with SHA-256
Signature: <i>(number of 128 bytes)</i>

The TrustMePlease certificate contains its public key, 17, and its RSA modulus (*a 128-byte number*).

Questions

1. Write the operations that a user must do to validate the digital certificate's signature.
2. ¿Would the certificate validate today?
3. Once the user has validated the University's signature, in order to communicate securely, how could the Simplified-SSL protocol continue? (This protocol is made up, the purpose of this question is to explain how a secure communication would be established once the signature has been verified)
4. How does the user know if he/she can trust the TrustMePlease authority? Are the key sizes and algorithms used adequate? Would you make any improvements to the protocol, and why?
5. What should change in the above certificate if the certificate authority decides to use DSA?